UP LT GRADE · COMPUTER SCIENCE MAINS

Software Engineering — High-Probability Numerical Questions with Full Solutions

Selected for Likelihood of Appearing, Based on the Confirmed Exam Pattern

Coverage: Effort estimation (COCOMO), code complexity, function points, reliability metrics, project scheduling (CPM), risk analysis, and earned value analysis.

Each question is tagged with its appearance likelihood, with the complete formula and step-by-step substitution shown.

Practice set · step-by-step solutions · exam-style presentation

Table of Contents

TYPE 1 · EFFORT ESTIMATION

Q1. Basic COCOMO — Effort & Development Time

Very High Chance

Question

A project is estimated at 50 KLOC and classified as an Organic type project. Using the Basic COCOMO model, calculate (a) the Effort in person-months, and (b) the Development Time in months.

(a) Effort Calculation

Formula (Organic: a=2.4, b=1.05)

Effort = a × $(KLOC)^b$ person-months

Effort = 2.4 × $(50)^{1.05}$

$50^{1.05}$ ≈ 60.8

Effort = 2.4 × 60.8 ≈ **146 person-months**

(b) Development Time Calculation

Formula (Organic: c=2.5, d=0.38)

Development Time = c × $(Effort)^d$ months

Tdev = 2.5 × $(146)^{0.38}$

$146^{0.38}$ ≈ 6.65

Tdev = 2.5 × 6.65 ≈ **16.6 ≈ 17 months**

Final Answer

Effort ≈ **146 person-months**, Development Time ≈ **17 months**

Why This Is Very High-Probability

COCOMO is the single most quoted estimation model in Software Engineering — always memorize BOTH formulas and the a,b,c,d constants for all three project types (Organic/Semi-Detached/Embedded), since the question may specify any of the three.

TYPE 2 · CODE COMPLEXITY

Q2. Cyclomatic Complexity from Control Flow Graph

Very High Chance

Question

A program's control flow graph has 11 edges (E), 9 nodes (N), and 1 connected component (P). Calculate the Cyclomatic Complexity, and state what the result means for testing.

Apply the Formula

Formula

$V(G) = E - N + 2P$

$V(G) = 11 - 9 + 2(1) = 11 - 9 + 2 = \mathbf{4}$

Final Answer

Cyclomatic Complexity = **4**. This means there are 4 linearly independent paths through the program, so a minimum of **4 test cases** are needed to achieve full Basis Path Coverage (testing every independent path at least once).

Why This Is Very High-Probability

Cyclomatic Complexity numericals almost always ask you to first DRAW the control flow graph from given code/pseudocode, THEN count E and N yourself — practice sketching flow graphs for simple if/while/for constructs, not just applying the formula to given E,N values.

TYPE 3 · SIZE ESTIMATION

Q3. Function Point Analysis

High Chance

Question

A system has the following (average complexity) counts: External Inputs (EI) = 4, External Outputs (EO) = 3, External Inquiries (EQ) = 2, Internal Logical Files (ILF) = 1, External Interface Files (EIF) = 2. The sum of the 14 General System Characteristics (GSC) ratings is 25. Calculate the Function Points (FP).

Step 1: Calculate Unadjusted Function Points (UFP)

Average Complexity Weights

EI = 4, EO = 5, EQ = 4, ILF = 10, EIF = 7 (weight per item, average complexity)

UFP = (4×4) + (3×5) + (2×4) + (1×10) + (2×7)
= 16 + 15 + 8 + 10 + 14 = **63**

Step 2: Calculate Value Adjustment Factor (VAF)

Formula

$VAF = 0.65 + (0.01 \times \Sigma F_i)$

VAF = 0.65 + (0.01 × 25) = 0.65 + 0.25 = **0.90**

Step 3: Calculate Final Function Points

Formula

FP = UFP × VAF

FP = 63 × 0.90 = **56.7 ≈ 57 Function Points**

Final Answer

Function Points ≈ **57**

TYPE 4 · RELIABILITY METRICS

Q4. MTTF, MTTR, MTBF & Availability

High Chance

Question

A system was observed for 1000 hours of total operating time, during which 5 failures occurred, with a total repair time of 20 hours across all failures. Calculate MTTF, MTTR, MTBF, and Availability.

Step 1: MTTF (Mean Time To Failure)

Formula

MTTF = (Total Operating Time − Total Repair Time) / Number of Failures

MTTF = (1000 − 20) / 5 = 980 / 5 = **196 hours**

Step 2: MTTR (Mean Time To Repair)

MTTR = Total Repair Time / Number of Failures = 20 / 5 = **4 hours**

Step 3: MTBF (Mean Time Between Failures)

Formula

MTBF = MTTF + MTTR

MTBF = 196 + 4 = **200 hours**

Step 4: Availability

Formula

Availability (%) = MTTF / (MTTF + MTTR) × 100

Availability = 196 / 200 × 100 = **98%**

Final Answer

MTTF = 196 hrs, MTTR = 4 hrs, MTBF = 200 hrs, Availability = **98%**

TYPE 5 • QUALITY METRICS

Q5. Defect Density Calculation

Moderate Chance

Question

A software module contains 20,000 lines of code (20 KLOC). During testing, 45 defects were found. Calculate the Defect Density.

Apply the Formula

Formula

Defect Density = Total Number of Defects / Size (in KLOC)

Defect Density = 45 / 20 = **2.25 defects per KLOC**

Final Answer

Defect Density = **2.25 defects/KLOC** — this metric is often used to compare quality across modules or track improvement across releases (lower is better).

TYPE 6 · TESTING METRICS

Q6. Statement Coverage Calculation

Moderate Chance

Question

A program has 150 executable statements in total. A test suite, when executed, covers 135 distinct statements. Calculate the Statement Coverage percentage achieved.

Apply the Formula

Formula

Statement Coverage (%) = (Statements Executed / Total Statements) × 100

Statement Coverage = (135 / 150) × 100 = **90%**

Final Answer

Statement Coverage = **90%** — meaning 15 statements (10%) were never executed by any test case and remain unverified, a White-Box testing adequacy metric.

TYPE 7 • PROJECT SCHEDULING

Q7. Critical Path Method (CPM)

High Chance

Question

A project has activities: A (3 days, no dependency), B (4 days, depends on A), C (2 days, depends on A), D (5 days, depends on B and C), E (3 days, depends on D). Find the Critical Path and the minimum project duration.

Step 1: List All Possible Paths from Start to End

Path 1: A → B → D → E = 3+4+5+3 = **15 days**
Path 2: A → C → D → E = 3+2+5+3 = **13 days**

Step 2: Identify the Critical Path (Longest Path)

Rule

The Critical Path is the LONGEST path through the network — it determines the minimum possible project duration, since every activity on it must finish on time or the whole project is delayed.

Final Answer

Critical Path = **A → B → D → E**, Minimum Project Duration = **15 days**

Path A-C-D-E has 2 days of "float" (slack) — activity C could be delayed by up to 2 days without affecting the overall project finish date.

TYPE 8 • RISK MANAGEMENT

Q8. Risk Exposure Calculation

Moderate Chance

Question

A project risk has a 30% probability of occurring. If it does occur, the estimated financial loss is ₹2,00,000. Calculate the Risk Exposure (RE).

Apply the Formula

Formula

Risk Exposure (RE) = Probability of Occurrence × Potential Loss

RE = 0.30 × ₹2,00,000 = **₹60,000**

Final Answer

Risk Exposure = **₹60,000** — this quantified value helps project managers PRIORITIZE which risks deserve the most mitigation effort/budget, by comparing RE values across multiple identified risks.

TYPE 9 · PROJECT TRACKING

Q9. Earned Value Analysis (SPI & CPI)

Moderate Chance

Question

At a project checkpoint: Planned Value (PV) = ₹1,00,000, Earned Value (EV) = ₹80,000, Actual Cost (AC) = ₹90,000. Calculate the Schedule Performance Index (SPI) and Cost Performance Index (CPI), and interpret the results.

Step 1: Schedule Performance Index (SPI)

Formula

SPI = EV / PV

SPI = 80,000 / 1,00,000 = **0.8**

Step 2: Cost Performance Index (CPI)

Formula

CPI = EV / AC

CPI = 80,000 / 90,000 = **0.889**

Final Answer

SPI = **0.8** (SPI < 1 means the project is BEHIND SCHEDULE — only 80% of planned work has actually been completed).
CPI = **0.889** (CPI < 1 means the project is OVER BUDGET — it's costing more than the value of work actually completed).

Key Interpretation Rule

For both SPI and CPI: a value of exactly 1 = on target, >1 = ahead/under-budget (good), <1 = behind/over-budget (concerning). Always state this interpretation, not just the numeric value.

TYPE 10 • RELIABILITY MODELING

Q10. Software Reliability (Exponential Model)

Moderate Chance

Question

A software system has a constant failure rate $\lambda = 0.02$ failures/hour. Using the Exponential Reliability Model, calculate the reliability $R(t)$ for a mission time of $t = 10$ hours.

Apply the Exponential Reliability Formula

Formula

$R(t) = e^{-\lambda t}$

$R(10) = e^{-(0.02)(10)} = e^{-0.2}$

$e^{-0.2} \approx$ **0.8187**

Final Answer

$R(10) \approx$ **0.819 (81.9%)** — there is about an 81.9% probability the software will run for 10 hours without failure, given its constant failure rate.

— End of Practice Set —